

Comprehensive Final Project Report

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Comprehensive Final Project Report

Project WikiMove – Migrating a Company Knowledge Base to a New Platform

Complete Academic Software Engineering & Project Management Portfolio

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Abstract & Executive Summary

Over 11 years of continuous agile software engineering, an enterprise software organization accumulated 14,000 internal wiki pages. Because no formal lifecycle governance or content custody was established, the documentation repository decayed: an internal audit revealed that approximately 50% of the entire knowledge base (7,000 pages) was outdated, redundant, or orphaned. Simultaneously, an immovable operational boundary emerged: the legacy platform software licence expires in 9 months (~39 weeks).

The Chief Technology Officer (CTO) mandated Project WikiMove to transition valuable technical documentation to a modern knowledge platform, permanently decommission obsolete material, and establish mandatory content custody. The project constraints were uniquely demanding: only a single part-time Technical Writer (3 person-days per week) was dedicated to lead the effort, while 12 engineering teams contributed 1 person-day per week each, establishing a fixed weekly capacity ceiling of 15 person-days per week.

This comprehensive final report presents the complete software engineering and project management deliverables for WikiMove. Grounded in empirical mathematical modeling:

1. Total Effort: Triage ($\$14,000 / 40 = 350 \text{ p-d}\$$) + Migration ($\$7,000 / 25 = 280 \text{ p-d}\$$) = 630 person-days.
2. Project Duration: $\$630 / 15 = \mathbf{42.0 \text{ weeks}\$}$.
3. Licence Overrun Conflict: 42 weeks required vs. 39 weeks available = +3.0 weeks schedule deficit, resolved through proactive management contingencies.
4. Risk Profile: R2 (Teams do not review, $\$E=4.2\$\$) ranked #1; R1 (Licence expiration, $\$E=3.6\$\$) ranked #2; R3 (Broken links, $\$E=1.6\$\$) ranked #3; R4 (No owner, $\$E=0.5\$\$) ranked #4.

The project encompasses a complete IEEE 830 SRS, BRD, UML models (7 diagrams), 6 responsive UI mockups, testing plans, RMMM frameworks, and a sustainable post-migration governance architecture.

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1. Problem Statement & Operational Genesis

In fast-paced software organizations, documentation often lags behind production code. Over 11 years, the company's internal wiki suffered from classic documentation entropy:

- 14,000 Unstructured Pages: Accumulation of sprint notes, obsolete API drafts, duplicates, and personal checklists.
 - 50% Decay Factor: 7,000 pages contained obsolete configurations, invalid credentials, dead URLs, and deprecated architectures.
 - Departed Custodians: Engineers transitioned or left the firm, leaving thousands of pages orphaned without designated maintenance.
 - The Licence Imposition: The legacy platform licence expires in 9 months (~39 weeks), rendering renewal financially and strategically unviable.
 - Resource Starvation: Engineering teams had zero dedicated time; project ownership defaulted to a single part-time technical writer (3 p-d/wk) and an enforced 1 p-d/wk per squad across 12 teams.
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2. Project Objectives & Academic Boundaries

2.1 Project Objectives

1. OBJ-01: Full inventory and triage of all 14,000 legacy wiki pages.
2. OBJ-02: Quality-gated migration of exactly 7,000 kept pages (50% retention).
3. OBJ-03: Enforce a strict "Zero Orphan Policy" binding every kept page to an active named owner.
4. OBJ-04: Ensure 0.0% broken internal links (NFR-01) and sub-second search discoverability (NFR-02).
5. OBJ-05: Resolve the 3-week schedule gap between baseline duration (42 weeks) and licence expiration (39 weeks).
6. OBJ-06: Establish an automated 90-day periodic revalidation cycle to prevent future decay.

2.2 Academic Scope Boundary

Consistent with teacher instructions for Software Engineering and Project Management:

- No production backend, cloud microservices, or database engines are deployed.
 - No Python scripts or runtime dependencies are utilized.
 - The UI is implemented as a lightweight, static prototype using pure HTML, CSS, and minimal JavaScript.
 - All documentation is provided in professional, editable DOCX and PDF formats.
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3. Existing Situation vs. Proposed System

Dimension	Legacy Situation	WikiMove Proposed Solution
Total Scope	14,000 unmanaged legacy pages	14,000 pages triaged; 7,000 clean pages migrated
Content Accuracy	~50% outdated, misleading, or redundant	100% verified, production-aligned documentation
Page Custody	Thousands of orphaned pages	Mandatory named human owner bound to every page
Governance	Unmanaged growth, zero periodic reviews	Automated 90-day periodic recertification
Licence Deadline	Expiring in 9 months (~39 weeks)	Structured offboarding within available window
Search & Links	Broken relative URLs, unindexed tags	100% rewritten URLs, certified search indexing

4. BRD Summary

The Business Requirements Document (WikiMove_BRD) baselines 12 key business requirements (BR-01 to BR-12) and 8 strict business rules:

- BRUL-01 (Orphan Prohibition): No page can migrate without an active, validated named content owner.
- BRUL-04 & BRUL-05 (Velocity Benchmarks): Triage fixed at 40 pages/person-day; migration fixed at 25 pages/person-day.
- BRUL-06 & BRUL-07 (Capacity & Retention): 50% retention (7,000 pages); weekly capacity capped at 15 person-days/week.

5. IEEE 830 SRS Summary

The Software Requirements Specification (WikiMove_SRS) defines 15 functional requirements (FR-01 to FR-15) and 10 non-functional requirements (NFR-01 to NFR-10). It provides bidirectional traceability through an IEEE-compliant Requirements Traceability Matrix (RTM) linking all functional specifications to verification test cases.

6. Functional & Non-Functional Requirements Architecture

- Functional Scope (FR-01..FR-15): Page inventory ingestion, review UI, Keep/Archive/Delete classification engine, named owner binding, migration batch planning (25 p/d), link validation

crawler, search discoverability tester, 42-week schedule tracking, risk exposure ranking, and 90-day governance rules.

- Key NFRs:
 - NFR-01 (Link Integrity): 0.0% broken internal links across 7,000 migrated pages.
 - NFR-02 (Search Performance): Search query response latency < 1.0 second.
 - NFR-04 (Usability): Mean review duration < 12 minutes/page (supporting 40 p/d triage quota).
 - NFR-08 (Auditability): 100% captured event history for all triage and owner changes.
-

7. UML System Modeling (7 Core Diagrams)

The proposed WikiMove platform is formally specified across 7 vector UML and system diagrams:

1. System Context Diagram (Context_Diagram.svg): Establishes external actors (CTO, Technical Writer, 12 Engineering Teams, QA) interacting with the WikiMove system boundary between legacy and target platforms.

2. Use Case Diagram (Use_Case_Diagram.svg): Maps 6 actors to 11 functional use cases (View Dashboard, Review Page, Classify Page, Keep/Archive/Delete, Assign Owner, Plan Migration, Track Risks, etc.).

3. Class Diagram (Class_Diagram.svg): Models core entity structures including WikiPage, TriageDecision, Owner, EngineeringTeam, MigrationBatch, ValidationResult, RiskItem, and ProjectSchedule.

4. Sequence Diagram (Sequence_Diagram.svg): Illustrates the message interchange across Reviewer, UI, TriageManager, Catalog, and Target Platform for triage and migration.

5. Activity Diagram (Activity_Diagram.svg): Specifies decision logic, condition branching (current? historical value? owner assigned?), and batch queues.

6. State Machine Diagram (State_Diagram.svg): Tracks page lifecycle states from Discovered to UnderReview, Deleted, Archived, OwnerAssigned, Migrated, Validated, and ActiveGoverned.

7. Logical Architecture Diagram (Architecture_Diagram.svg): Presents a 4-tier layered architectural model separating UI prototypes, management services, data models, and external platforms.

8. Simple Logical System Architecture

%
% 1. PRESENTATION LAYER:
Responsive Static Prototype
(6 HTML/CSS Screens) %
% [Screen 1: Dashboard]
[Screen 2: Triage] [Screen
3: Review] %
% [Screen 4: Migration]
[Screen 5: Ownership]
[Screen 6: Risks & Status] %
%

% HTTP / DOM Events
%
% 2. APPLICATION MANAGEMENT
LAYER (Proposed Workflow
Specifications) %
% • Inventory & Triage (40
p/d) • Ownership Custody
Registry %
% • Migration Batching (25
p/d) • Schedule & Risk
Guard (42 vs 39 Wk)%
%

% Data Abstraction
%
% 3. DATA & TRACEABILITY
MODEL LAYER (Proposed
Schemas) %
% • 14,000 Legacy Page
Records • RTM
Traceability
Schema %
% • Triage & Decision Audit
Logs • Defect Density &
DRE Logs %
%

% Integration Interface
%
% 4. EXTERNAL ENTERPRISE PLA
TFORMS

%
% • Legacy Wiki Platform
(Exp: ~39 Wks) • Target
Knowledge Platform %
% %

9. UI Design & Prototype Specification (6 Screens)

The user interface is designed as an accessible, clean academic prototype using HTML5, CSS3, and minimal JavaScript:

Dashboard (dashboard.html): Metrics for 14,000 total, 5,600 reviewed, 7,000 target kept, 7,000 outdated, progress bars, licence timer, and top risk card.

- Screen 2 — Page Triage Queue (triage.html): Data table showing Page ID, title, team, last updated date, suggested action, and simulated Keep, Archive, and Delete buttons.

- Screen 3 — Detailed Page Review (review.html): Split-view content preview, revision history, reviewer notes, and mandatory named owner dropdown.

- Screen 4 — Migration Dashboard (migration.html): 7,000 kept pages progress bar (2,100 completed, 420 validation pending, 4,460 queued), batch breakdown table.

- Screen 5 — Content Ownership Management (ownership.html): Registry of active content custodians, next review dates, and orphan escalation alerts.

- Screen 6 — Risks & Project Status (risks.html): Quantitative risk table (R1-R4), exposure ranking, and mathematical schedule gap analysis table.

10. Project Management Plan & WBS

The project is decomposed into 9 distinct phases spanning 42 weeks:

- Phase 1: Initiation (Weeks 1–2)
- Phase 2: Requirements & Architecture (Weeks 3–4)
- Phase 3: Wiki Inventory (Weeks 5–6)
- Phase 4: Content Triage (Weeks 7–22)
- Phase 5: Migration Preparation (Weeks 20–22)
- Phase 6: Content Migration (Weeks 23–36)
- Phase 7: Testing & QA Validation (Weeks 34–38)
- Phase 8: Ownership & Governance (Weeks 37–40)
- Phase 9: Closure & Handover (Weeks 41–42)

11. Mathematical Estimation & Schedule Analysis

Every project parameter is derived

from empirical case-study formulas:

$$\begin{aligned} \text{Total Pages } (N) &= 14,000 \\ \text{Outdated Ratio} &= 50\% \\ \implies \text{Kept Pages } (N_{\text{kept}}) &= 14,000 \times 50\% = \\ &= \mathbf{7,000 \text{ pages}} \\ \text{Triage Effort } (E_{\text{triage}}) &= \frac{14,000}{40 \text{ p/d}} = \\ &= \mathbf{350 \text{ person-days}} \\ \text{Migration Effort } (E_{\text{migration}}) &= \frac{7,000}{25 \text{ p/d}} = \\ &= \mathbf{280 \text{ person-days}} \\ \text{Total Effort } (E_{\text{total}}) &= 350 + 280 = \mathbf{630 \text{ person-days}} \\ \text{Weekly Capacity } (C) &= 3 \text{ (TW)} + 12 \text{ (12 Teams)} \\ &= \mathbf{15 \text{ person-days / week}} \\ \text{Estimated Duration } (D) &= \frac{630}{15} = \\ &= \mathbf{42.0 \text{ weeks}} \\ \text{Licence Expiry Period } (T_{\text{licence}}) &= 9 \text{ months} \times 4.333 \text{ wks/mo} = \\ &= \mathbf{39.0 \text{ weeks}} \\ \text{Schedule Gap } (\Delta T) &= 42.0 - 39.0 = \\ &= \mathbf{+3.0 \text{ weeks (Schedule Overrun Deficit)}} \end{aligned}$$

12. Gantt Chart & Milestone Schedule (42 vs. 39 Weeks)

The Gantt schedule explicitly marks the conflict at Week 39, where the legacy licence expires while baseline activities extend through Week 42.

13. Critical Path Method (CPM) Analysis

$$\begin{aligned} \text{Critical Path: Requirements} &\rightarrow \text{Inventory} \\ &\rightarrow \text{Triage (16 wks)} \\ &\rightarrow \text{Migration (14 wks)} \\ &\rightarrow \text{Testing} \rightarrow \text{Closure} \end{aligned}$$

Total Critical Path length = 42.0 weeks (Zero Float). To mitigate the 3-

week overrun, three management contingencies are established:

1. Commercial 30-day bridging licence extension through Week 43.
2. Capacity compression increasing TW allocation from 3 to 5 p-d/week.
3. Phased cutover prioritizing top 5,000 core pages before Week 39.

14. Testing & Quality Assurance Documentation

- Functional Test Cases: 15 comprehensive test cases (TC-01 to TC-15) documented with Status = Planned and Actual Result = To be executed during implementation.
- Boundary Value Analysis (BVA): Evaluates boundaries for inventory (0, 1, 14000, 14001), triage rate (40 p/d), batch size (500 pg), and search latency (< 1.0s).
- Decision Tables: Complete triage decision rules and migration QA certification rules.
- Defect Removal Efficiency (DRE): $\text{DRE} = \frac{E}{E + D} \times 100\%$ (Target benchmark: $\geq 95.0\%$).
- Defect Density: Normalized metric per 1,000 tested pages (Target ceiling: $\leq 8.0 \text{ defects / KPages}$).

15. Quantitative Risk Management & RMMM Plans

Risk Exposure formula $E = P \times I$ applied to all 4 case study risks:

1. Risk 2 (Teams do not review pages): $P=0.6$, $I=7$ implies $E=4.2$ (Rank 1 — Highest Risk). Mitigated via CTO executive mandate, 1 p-d/wk sprint tracking, and TW pre-filtering.
2. Risk 1 (Licence expires before completion): $P=0.4$, $I=9$ implies $E=3.6$ (Rank 2 — High Risk). Mitigated via 30-day bridging licence, critical path tracking, and core page staging.
3. Risk 3 (Broken links after migration): $P=0.4$, $I=4$ implies $E=1.6$ (Rank 3 — Moderate Risk). Mitigated via automated regex URL rewrite tables and crawler verification.
4. Risk 4 (No owner after migration): $P=0.5$, $I=1$ implies

\mathbf{E=0.5}\\$ (Rank 4 — Low Risk). Mitigated via system-enforced Zero Orphan Policy and 90-day review cycles.

16. Post-Migration Governance Framework

To permanently eliminate knowledge entropy, WikiMove establishes a 5-pillar governance model:

1. Mandatory Named Custodian: 100% of pages mapped to active employees.
 2. Automated 90-Day Periodic Review: Automated prompts requiring owner recertification.
 3. 14-Day Escalation SLA: Unrevalidated pages escalate to domain Engineering Leads.
 4. Semi-Annual Archive Sweep: Inactive pages retired to cold storage.
 5. Quarterly Broken Link Audits: Scheduled background crawlers verify all intra-wiki and external links.
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17. Project Closure & Handover State

- Planned Closure State: All 14,000 legacy pages triaged; 7,000 kept pages migrated and verified; legacy wiki platform decommissioned; target platform live under active governance.
 - Handover Package: Wiki inventory registry, URL rewrite map, ownership directory, and QA crawler scripts transferred to the permanent technical writing team.
-

18. Lessons Learned & Academic Post-Mortem

1. Documentation Requires Formal Ownership: Without named custody, technical knowledge bases inevitably decay into unmaintainable repositories.
2. Mathematical Estimation Exposes Latent Conflicts: Calculating effort against fixed capacity early revealed the 3-week schedule deficit, allowing proactive risk mitigation rather than reactive crisis management.
3. Cross-Team Governance Requires Executive Backing: Relying on voluntary engineering team review fails; institutionalizing review quotas

in sprint capacity is critical.

19. Conclusion

Project WikiMove provides an academically rigorous, mathematically consistent, and professional software engineering and project management portfolio. By addressing legacy content decay through structured triage, quality-gated migration, quantitative risk management, and long-term governance, the proposed system ensures that the organization successfully offboards its legacy platform and establishes a sustainable enterprise knowledge repository.

20. References & Appendices

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- Pressman, R. S., & Maxim, B. R.: Software Engineering: A Practitioner's Approach.
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